

TOPIC 2.3

How is energy involved in chemical processes?

Key Concepts

- Matter and energy interact in physical and chemical changes.
- There is a transfer of energy between chemical reactions and the surroundings.

Curricular Competencies

- Collaboratively and individually plan, select, and use appropriate investigation methods.
- Use knowledge of scientific concepts to draw conclusions consistent with evidence.
- Evaluate methods and experimental conditions.
- Formulate physical or mental theoretical models to describe a phenomenon.

British Columbia's Gold Rush Trail Sled Dog Mail Run takes place for three days every January. Mushers are sworn in as official mail carriers and carry Canada Post mail in their dog sleds to post offices along the route, in Quesnel, Wells, and Barkerville. Although once a race, this event is now about community participation. In addition to the dogs, skiers, snowshoers, runners, and kick-sledders are welcome. All participants (dogs and humans) depend on chemical reactions in their body cells for the energy they need. This energy is released as chemical bonds holding together nutrient compounds are broken and then reformed to produce new compounds.



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Identifying Preconceptions** What does the term *energy* mean? Give some examples of how you have heard the word used both inside and outside the science classroom.
2. **Questioning and Planning** The process that releases energy from food in the cells of living things is called cellular respiration. The first scientist to study this process in depth was Antoine Lavoisier. He noticed that it is similar to another chemical reaction that he had been studying closely: combustion. Write a word or chemical equation for each of these processes to see why Lavoisier might have been interested to study both. What kinds of questions do you think he might have asked about them? What kinds of experiments might he have planned as a result?
3. **Considering First Peoples' Perspectives** In many First Peoples societies, people sometimes place a small portion of their meal in the flames of a fire. For some, this is a way of thanking the land for the food, and for others they are feeding their ancestors. How do these ceremonies demonstrate a holistic view of the energy of fire?



Key Terms

There are two key terms that are highlighted in bold type in this topic:

- exothermic reaction
- endothermic reaction

Flip through the pages of this Topic to find these terms. Add them to your class Word Wall along with their meaning. Add other terms that you think are important and want to remember.

CONCEPT 1

Matter and energy interact in physical and chemical changes.

Activity

Changes and Energy

Your teacher will provide you and a partner with a piece of ice. What happens to the ice as it sits at room temperature? How is energy involved in this process? What happens if the ice is in your hand? Is the amount of energy in your hand increasing or decreasing? Support your answers with evidence from the activity.



Figure 2.16 Many common events at home require an input of energy.

Matter and energy are continually interacting in the world around us. Cooking food is just one common example. To boil the water and make the toast in **Figure 2.16**, energy is needed. The temperature of the water increases as energy is transferred from the stove burner, and the changes to the bread rely on the addition of energy from a toaster.

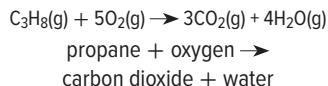
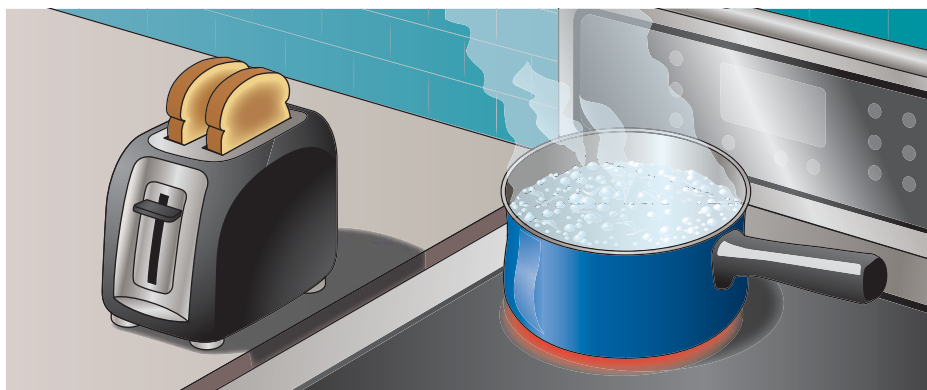


Figure 2.17 A lighter provides energy to start the reaction between oxygen and propane. **Predicting:** If energy input was not needed to light this torch, what would happen when propane was released from the tank?

Getting Things Started

For any chemical change to occur, the reactants must collide with enough energy to begin to break the bonds in the reactants. This minimum amount of energy needed for a reaction to occur is called the *activation energy*. It is often useful to think of the activation energy as a barrier or “hill” that needs to be overcome for a reaction to happen.

Many chemical reactions require an initial input of energy—the reactants will not react by simply mixing them together. For example, lighting a propane torch requires a spark or lighter (**Figure 2.17**). This is the energy input needed for the chemical reaction between oxygen (in the air) and the propane gas. The lighter provides a few molecules of oxygen and propane with enough energy to overcome the energy barrier. In the case of this reaction, a large amount of energy is released—that’s what the flame and heat to weld material are. Once the propane is ignited, it will continue to burn because the energy released during the reaction provides the energy for other molecules to overcome the energy barrier to react.

How does energy change help to solve crimes?

What's the Issue?

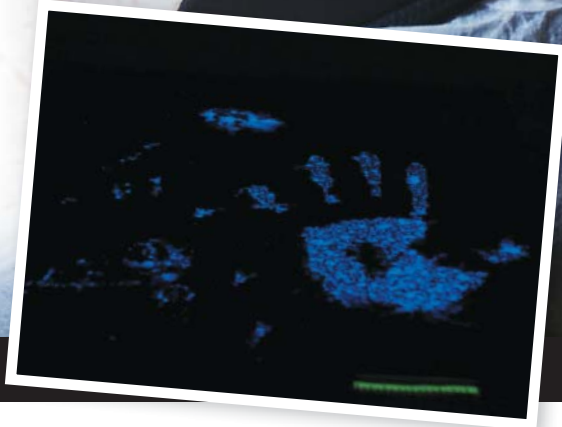
A forensics investigator is testing a pair of garage coveralls to find out if the red staining on the fabric is blood. After spraying the coveralls with a solution containing a compound called luminol, a glowing blue-green light shows up—suggesting foul play.

Energy Helps to Solve Crimes

Luminol is a solid that is typically dissolved in a solution of hydrogen peroxide that contains a source of hydroxide ions, such as KOH(aq) . The solution can be sprayed onto an object or an area to test for blood.

Luminol reacts with iron from hemoglobin molecules in blood to form a compound that is in an excited state. This means that an electron in each luminol molecule has absorbed energy and has been excited to a higher energy level. When the electron drops back to its original energy level, energy is released as light. This light can be seen easily in a darkened room and lasts for about 30 seconds. Luminol is so sensitive to the iron in blood that blood can be detected even when it cannot be seen.

A positive test does not always mean blood is present. Luminol reacts with other compounds that contain iron, as well as with copper and nickel. Bleach also reacts with luminol. Saliva and even horseradish can also cause luminol to emit light.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. What happens when luminol interacts with blood?
2. Does a positive luminol test provide conclusive evidence that a stain is blood? Explain.
3. What is DNA? Why do you think DNA testing of the blood is the next step in the investigation? Why would investigators do the test with luminol first?

CONCEPT 2

Energy is transferred between chemical reactions and the surroundings.

Activity

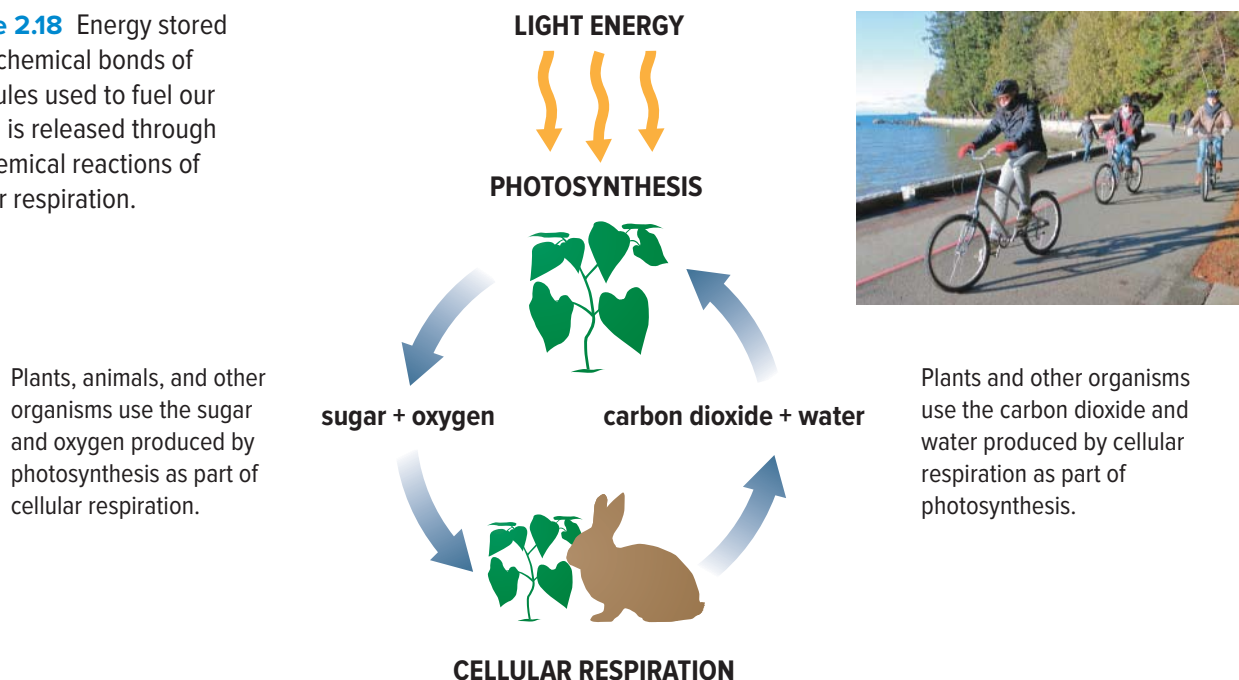
Where Did the Energy Go?



1. Read the procedure and make a table to record your observations.
2. Place 5 mL of vinegar into a large test tube that is in a test tube rack. Using a thermometer, record the temperature of the vinegar. Leave the thermometer in the test tube.
3. Add a small scoop (about 0.5 g) of baking soda to the test tube. Record the temperature every 5 seconds, until it no longer changes. Record any other observations. Draw a temperature versus time graph of your data. What is the overall temperature change? How would you describe the reaction in terms of energy changes? Was more energy involved in breaking bonds in this reaction or in forming bonds?

All chemical reactions are accompanied by changes in energy. These energy changes are crucial to life on Earth. For example, through photosynthesis green plants absorb energy from sunlight and store it in the chemical bonds of the sugar molecules they produce. Animals that eat the plants use the stored energy through the process of cellular respiration to fuel things such as growth and movement (**Figure 2.18**).

Figure 2.18 Energy stored in the chemical bonds of molecules used to fuel our bodies is released through the chemical reactions of cellular respiration.



The System and the Surroundings

Chemists think of energy changes in chemical reactions in terms of energy transfers between the *system* and the *surroundings*. The system is the materials involved in the chemical reaction and everything else in the universe is the surroundings. The *law of conservation of energy* states that the total energy of the universe is constant—energy cannot be created or destroyed. In terms of a chemical reaction, it means energy that leaves the system must enter the surroundings, and energy that enters the system must come from the surroundings (Figure 2.19).



Figure 2.19 The chemical reaction occurring in the flask is the system. The flask, the student, and the laboratory are part of the surroundings.

Exothermic and Endothermic Reactions

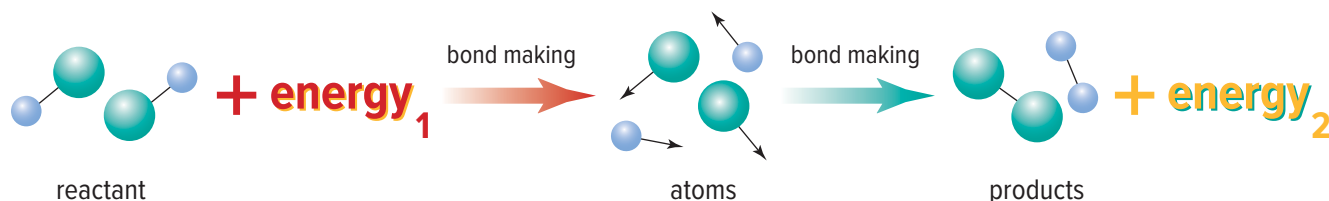
Recall that energy must be absorbed to break bonds and energy is released when bonds form. By comparing the total energy used when bonds in the reactants are broken with the total energy released when bonds in the products are formed, you can determine whether there is an overall release of energy or absorption of energy in a chemical reaction.

When, overall, more energy is released into the surroundings than absorbed by the system, the reaction is called an **exothermic reaction**. (The prefix *exo* means “external” and *thermic* refers to heat.) In exothermic reactions, there is more energy released from the formation of bonds than energy required to break the bonds. When there is an overall greater amount of energy absorbed by the system than released to the surroundings, the reaction is called an **endothermic reaction** (*endo* means “internal”). In endothermic reactions, there is more energy required to break the bonds than energy released from the formation of bonds (Figure 2.20).

exothermic reaction a chemical reaction in which there is net release of energy to the surroundings

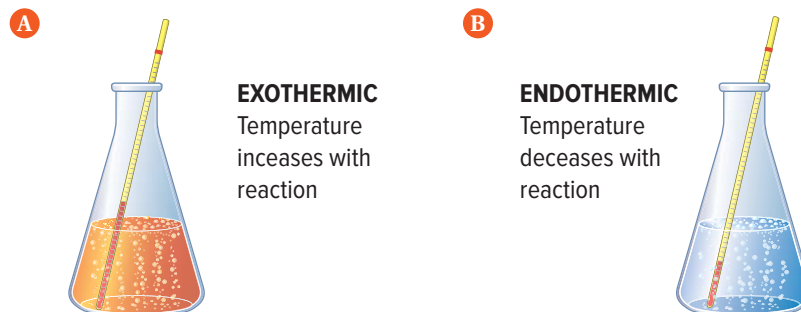
endothermic reaction a chemical reaction in which there is net absorption of energy from the surroundings

Figure 2.20 When the energy₂ term is greater than the energy₁ term, the overall process is exothermic. When energy₁ is greater than energy₂, the overall process is endothermic.



Connect to Investigation 2-D on page 150

Figure 2.21 Comparing the temperature of reactions occurring in solution.
Analyzing: How do the relative amounts of energy absorbed and released by the reactions compare?

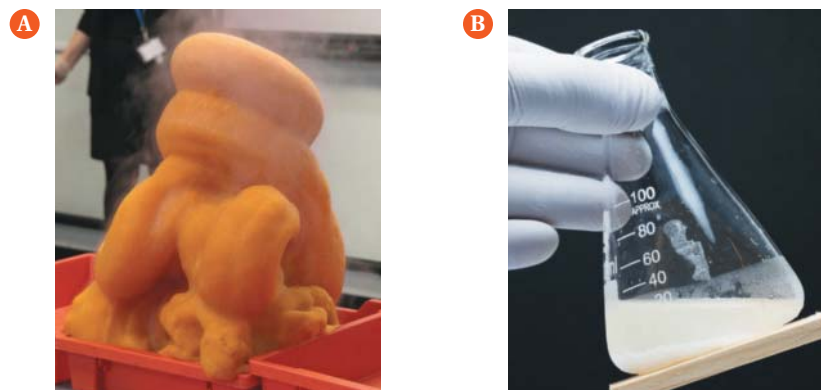


Connect to Investigation 2-E on page 152

Measuring Energy Changes

Energy changes in a reaction can be monitored by measuring change in temperature. For example, many chemical reactions occur in a solvent such as water. In **Figure 2.21A**, the temperature of the solution increases as energy released from the reaction (system) is transferred to the surroundings. In **Figure 2.21B**, the temperature of the solution decreases as energy is absorbed by the reaction from the surroundings. Sometimes, though, as in **Figure 2.22**, you don't need a thermometer to tell if a reaction is exothermic or endothermic.

Figure 2.22 Steamy foam in **A** results from a reaction of yeast, dish soap, and hydrogen peroxide. In **B**, the reaction of ammonium thiocyanate and barium hydroxide freezes the flask to the wooden base.
Analyzing: Which reaction is absorbing energy?



Activity

Warming Up

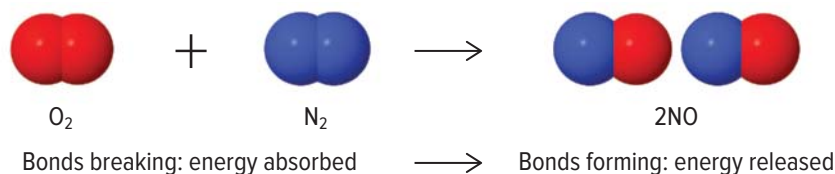


The reaction you will perform is $2\text{NaHCO}_3 + \text{CaCl}_2 \rightarrow \text{CaCO}_3 + \text{CO}_2 + 2\text{NaCl} + \text{H}_2\text{O}$

1. Read the procedure and make an appropriate table to record your observations.
2. Add 0.3 g of baking soda to 5 mL of water in a large test tube in a test tube rack. Record the temperature of the solution. Leave the thermometer in the test tube.
3. Add 0.5 g of CaCl_2 to the test tube. Record the temperature every 5 s until it stops changing. Also record other observations. Draw a temperature-time graph of your data. Is the reaction exothermic or endothermic? What is your evidence?

Representing Energy Changes

Consider the reaction below. For every molecule of nitrogen that reacts with a molecule of oxygen, two molecules of nitrogen monoxide are produced.

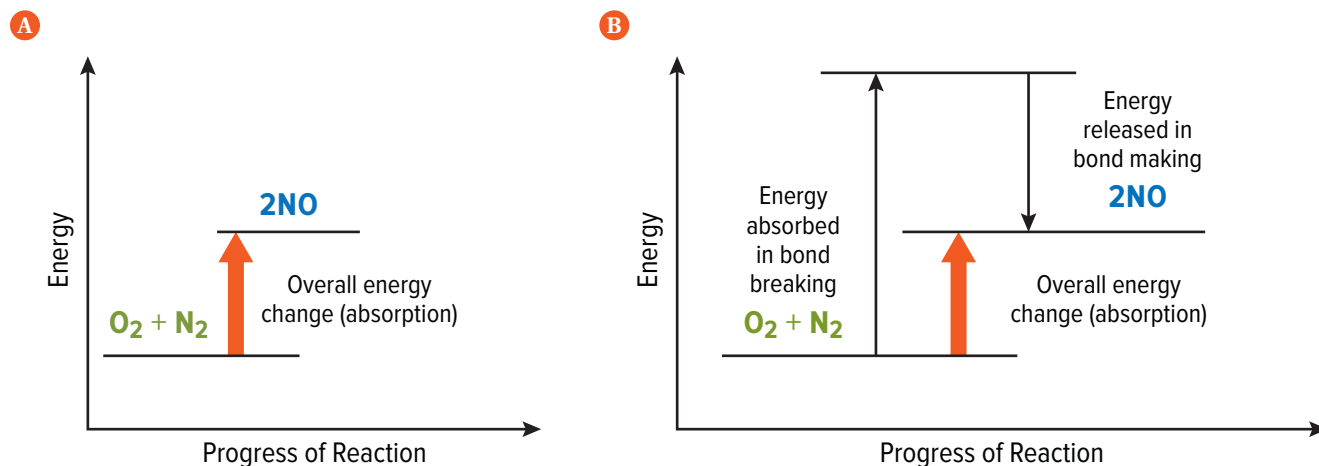


In this reaction, nitrogen-nitrogen bonds and oxygen-oxygen bonds are broken. The breaking of these bonds in each case absorbs energy. Nitrogen-oxygen bonds form, and the formation of these bonds releases energy. The total energy absorbed to break each nitrogen-nitrogen bond and oxygen-oxygen bond is more than the total energy released when nitrogen-oxygen bonds form. Therefore, there is an overall absorption of energy, and the reaction is endothermic.

Figure 2.23 shows how energy changes of a reaction can be represented. These are called energy-level diagrams, and they show the relative energy levels of the reactants and products in a reaction. For endothermic reactions, there is more energy required to break bonds than energy that is released when new bonds form. Therefore, there is an overall absorption of energy. This means that the energy of the products is greater than the energy of the reactants. For exothermic reactions, there is an overall release of energy, and the products are at a lower energy than the reactants.

Connect to Investigation 2-F on page 153

Figure 2.23 **A** This diagram shows the overall absorption of energy. Notice that the products have more energy than the reactants. **B** This diagram shows how the overall energy change is the net result of energy absorbed to break bonds and energy released in the forming of bonds.



Before you leave this page . . .

1. What is an endothermic reaction?
2. Describe the relative energies of reactants and products in an endothermic reaction.
3. If there is a decrease in energy of the system, what happens to the energy of the surroundings?

Why does it matter how quickly a reaction occurs?

What's the Issue?

Every chemical reaction happens at a certain rate under specific conditions. You can speed up or slow down the reaction by changing the conditions. For example, an increase in temperature, surface area, or concentration of a reactant will each cause a reaction to happen faster. So does the presence of a catalyst. (A catalyst makes a reaction happen faster without being used up in the reaction. For example, catalytic converters on vehicles contain metals that act as catalysts in reactions that convert harmful gases in exhaust to less harmful substances.)



An automobile catalytic converter (full view and cross-section)

Runaway Reactions

When a chemical reaction is “scaled up” from the laboratory to industry, extra caution is needed. A great deal of heat can be released during exothermic reactions. Inside a large reaction vessel, the safe removal of heat is essential, because it can cause a “runaway” reaction. Since an increase in temperature increases the rate of a reaction, heat released from a reaction can cause it to self-accelerate and become uncontrolled.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

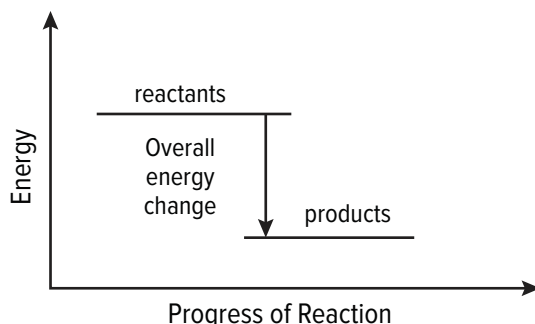
1. How fast or slow a reaction occurs is important. For example, the process of rusting is a chemical reaction that occurs very slowly, which is desirable. Give three examples of chemical reactions that demonstrate the importance of how fast a reaction occurs.
2. Chemical reactions depend on collisions between reactant atoms and molecules. Changing the number of collisions that happen in a certain amount of time will alter how quickly a reaction happens. Use this as a basis for developing an explanation for the effect of temperature, surface area, and amount of reactant on the speed of a reaction.
3. Your body cells contain many different catalysts called enzymes. What can you infer about the reactions in your body and the link between enzyme activity in the body and health?
4. In a group discuss the issue of industrial chemical reactions and possible safety concerns. What could be done to prevent runaway reactions? How can the surrounding environment be protected?

Check Your Understanding of Topic 2.3

QP Questioning and Predicting **PC** Planning and Conducting **PA** Processing and Analyzing **E** Evaluating
AI Applying and Innovating **C** Communicating

Understanding Key Ideas

- Describe an example of a physical change that is endothermic and a physical or chemical change that is exothermic. **PA**
- Compare the overall energy changes that occur in endothermic reactions with those that occur in exothermic reactions. How are the energies of bond formation and bond breaking involved? **PA E**
- Draw a sketch of the overall transfer of energy between the system and surroundings for an endothermic reaction. **C**
- An energy-level diagram is shown below.



- Does the diagram represent an exothermic or endothermic reaction? Explain. **PA**
- Draw a diagram that would represent a greater overall energy change. **C**

Connecting Ideas

- Students perform a chemical reaction in a glass test tube. They notice that the test tube feels cooler than it did before the reactants were added. **PC PA**
 - Did the students most likely perform an exothermic or endothermic reaction?
 - What data could the students collect to confirm the type of reaction?
 - Identify the system and the surroundings in this investigation.

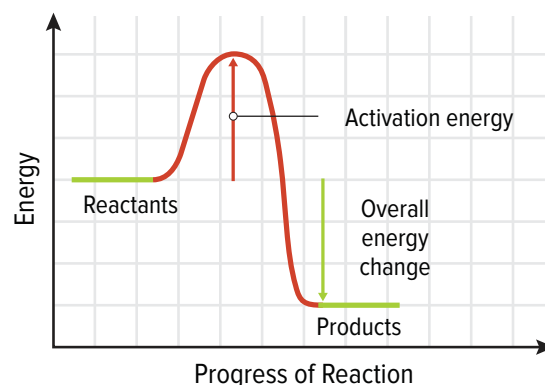
- Is melting an ice cube an endothermic or exothermic process? Explain. **QP AI**
- Although many individual reactions are part of photosynthesis and cellular respiration, the following chemical equations can be used to represent the overall processes.
 Photosynthesis:

$$6\text{H}_2\text{O} + 6\text{CO}_2 + \text{energy} \rightarrow 6\text{O}_2 + \text{C}_6\text{H}_{12}\text{O}_6$$
 Cellular respiration:

$$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy}$$
 Is photosynthesis an endothermic process or an exothermic process? What about cellular respiration? Explain your answers. **QP PA E**

Making New Connections

- The image below shows another way that energy changes in a chemical reaction can be represented. **E AI**



- What is activation energy? Why do you think it is represented as a “hill”?
- Describe the relative energy levels of the reactants and products.
- Does this diagram represent an exothermic or endothermic reaction? Explain your answer.
- Infer which are the most stable particles in the reaction. Which are the least stable? Justify your responses based on the energies of the particles.

Make a Difference

Ata Sina and the Shape-Shifting Paper

What do the Japanese art of origami and plastic polymer chemistry have in common? Ata Sina has an unusual answer to this question. As a graduate student at the University of British Columbia, he created a paper that folds, on its own, into intricate three-dimensional shapes like origami.

The paper is manufactured in several stages. First, a computer program is used to create carefully placed cuts and folds in the paper. Next, thermoplastic polymers are attached to the paper in certain locations. Then the paper spends 10 to 20 seconds in a 110°C oven. The thermoplastic polymers have been forced into the shape of long, stretched-out polymer chains. In the oven, the elongated polymers absorb energy, which increases the motion of their molecules. This increased motion allows the molecules to break out of their forced, elongated shape, which causes the thermoplastic to shrink. As it does so, it lifts the paper to form the three-dimensional shape.

When the thermoplastic shrinks, its strength increases as well. As a result, the structures

formed by the shape-shifting paper are strong, in addition to being light-weight and affordable. It also takes less energy to produce structures made of this material compared to plastic.

Evaluate and Communicate

1. Ata Sina's shape-shifting paper has multiple uses, such as an artistic material, sustainable packaging, or a children's toy. Create a marketing proposal for one of the above applications of shape-shifting paper or come up with your own. The proposal should also explain the chemistry behind the paper.
2. Ata Sina's shape-shifting paper allowed him to combine his interest in both art and science. What personal interests do you have that could be combined with science, and in what ways? Share your ideas with a partner or in a group. You might find that notions from various sources can be combined in unusual ways that inspire even more intriguing ideas.



Chemistry Connections



Questions

1. What other jobs and careers do you know or can you think of that involve working with chemicals or studying chemical reactions?
2. Research a job or career related to Unit 2 that interests you. Explain what attracted you to it. What kinds of things do you have to know, do, and understand for this job or career?

Skills and Strategies

- Processing and Analyzing
- Evaluating
- Communicating

Safety



- Wear safety goggles, gloves, and a lab coat.
- The materials can become quite hot. Avoid touching the hot liquid or glass. Use a thermal glove or tongs.

What You Need

Part 1:

- 35 mL of copper(II) chloride solution, 0.5 mol/L $\text{CuCl}_2(\text{aq})$
- two 250 mL beakers
- thermometer
- aluminum foil, 6 cm by 6 cm
- fluted filter paper for gravity filtration
- funnel
- ring clamp, retort stand, beaker, and stirring rod
- wash bottle, with tap water

Part 2:

- 40 mL tap water
- 250 mL Erlenmeyer flask
- thermometer
- 5 g sodium hydrogen carbonate (baking soda), NaHCO_3
- 5 g citric acid, $\text{C}_6\text{H}_8\text{O}_7$
- timer or stopwatch

Exothermic and Endothermic Reactions

Evidence that suggests a chemical change has occurred includes a change in temperature of the chemical reaction mixture. This temperature change is associated with energy changes that accompany chemical reactions. The overall energy change is the net result of energy that is absorbed to break chemical bonds and energy that is released when new chemical bonds form.

In this investigation, you will measure changes in temperature during a chemical reaction to determine if there is an overall release or absorption of energy. Based on your results, you will classify the reaction as endothermic or exothermic.

Question

How can exothermic and endothermic reactions be identified?

Procedure

Part 1:

1. Read Steps 2 to 7, and then design a table to record your observations.
2. Add 35 mL of copper(II) chloride solution to a beaker. Insert the thermometer. Loosely crumple the aluminum foil into a 1.5 cm diameter ball. Record the temperature of the solution to the nearest tenth of a degree and any other observations.

- Place the foil ball in the solution and monitor the chemical reaction that is occurring in the beaker. Record the temperature every 30 seconds and any other changes you observe as the reaction proceeds.
- Allow the reaction to cool. While you are doing that, watch as your teacher demonstrates how to use the filter paper and funnel to perform a gravity filtration so that you can separate the solid from the reaction mixture.
- Secure the funnel, with filter paper, to the retort stand by using a ring clamp. Make sure to place the beaker under the funnel to collect the liquid.
- Pour the reaction mixture through the filter paper. Pour slowly so it does not flow over the top of the funnel. Rinse the residue out of the first beaker with cold water using a wash bottle. Pour the rinse into the funnel.
- When all the fluid has passed through, carefully remove the filter with the solid you collected. Record your observations.
- Clean your work area, and dispose of materials according to your teacher's instructions.

Part 2:

- Read Steps 2 to 4, and then design a table to record your observations.
- Add 40 mL of water to an Erlenmeyer flask. Place the thermometer in the flask and record the temperature.
- Place 5 g baking soda in the Erlenmeyer flask. Swirl to dissolve the solid and record the temperature.
- Add 5 g citric acid to the flask, mix the contents, then record the temperature. Record the temperature every 15 seconds, using a stopwatch or timer to monitor the time. Also record any changes you observe as the reaction proceeds.
- Clean your work area and dispose of materials according to your teacher's instructions.

Process and Analyze

- Did a chemical reaction occur in Part 1? in Part 2? What observations support your answer?
- Graph the data you collected in Parts 1 and 2.
- What is the overall change in temperature for the reaction you performed in Part 1? What is the overall change in temperature for the reaction you performed in Part 2?

Conclude and Communicate

- Describe the reactions you performed in terms of the overall energy changes that occurred. How do the data you collected support these conclusions?
- Identify each reaction you performed as endothermic or exothermic. Describe evidence that supports your answer.
- The skeleton equation for the reaction between aluminum and copper(II) chloride is below.

$$\text{Al(s)} + \text{CuCl}_2\text{(aq)} \rightarrow \text{AlCl}_3\text{(aq)} + \text{Cu(s)}$$
 - Write the balanced chemical equation for this reaction.
 - What did you notice about the colour of the solution during the reaction? Propose an explanation.
 - What happened to the aluminum? Propose an explanation.
 - Based on your observations and explanations, describe what you think is happening to the atoms and ions in the reaction between aluminum and a solution of copper ions and chloride ions. What bonds are broken? What new bonds form?

Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing
- Evaluating

Safety



- Wear safety goggles, gloves, and a lab coat or apron.
- Handle chemicals safely. Avoid touching all reactants and products.
- Wash your hands thoroughly after completing this investigation.

What You Need

- test tubes or beakers
- thermometer
- water
- calcium chloride
- ammonium chloride
- sodium chloride
- other compounds your teacher may provide

What Makes the Best Hand Warmer?

Hand warmers can be slipped inside mitts to help combat cold conditions. One design has water on one side and on another side a substance that dissolves in the water when the packet is squeezed. In this investigation, the class will determine which chemical, when dissolved in water, would make the most effective component of a hand warmer.

Question

Which chemical makes the best component for a hand warmer?

Procedure

1. Your teacher will assign your group the chemicals to test. Design a procedure that identifies which of them, when dissolved in water, will make the most effective hand warmer. Your design should include the following considerations:
 - How quickly does the temperature change?
 - What is the maximum temperature that should be reached to ensure that skin is not harmed?
 - What volume is needed?
 - How expensive is it?
 - How safe is it for the person as well as the environment?
2. Write out the procedure for how you will test the chemicals. Include all safety precautions you will take, as well as clean-up and disposal instructions.
3. Have your teacher approve your procedure before proceeding.

Process and Analyze

1. What criteria did you use to assess which chemical was best? Explain why you chose them.
2. Summarize your data in a table.

Conclude and Communicate

3. As a class, share your results. Which chemical was identified as the most suitable for a hand warmer, and what was the rationale?
4. Identify each process you studied as exothermic or endothermic. Explain your answer.
5. Find out more about why the dissolving of ionic compounds in water can be exothermic or endothermic processes.

Skills and Strategies

- Questioning and Predicting
- Processing and Analyzing
- Evaluating
- Applying and Innovating
- Communicating

What You Need

- computer with Internet access

Chemical Reactions as Energy Sources

One way that society relies on chemical reactions is as a source of energy. Many of us use natural gas to heat our homes, gasoline to power vehicles, as well as more environmentally friendly alternatives such as hydrogen fuel cells and battery-powered cars. In this investigation, you will develop questions you have about one way our society produces energy by chemical reactions.

Procedure

1. Working in small groups, decide on what type of energy production from chemical reactions you want to investigate. This can include use of fossil fuels such as gasoline and propane at the individual level for heating, transportation, or cooking, as well as alternative energy sources such as hydrogen fuel cells.
2. Write out any questions your group has about the energy source and how chemistry is involved.
3. Decide which question or questions you will investigate and plan how you will answer them. Your investigation should include addressing the following:
 - What is the energy used for?
 - What is the chemical reaction and balanced chemical equation that represents it? Include a description of what is occurring to the atoms involved and what bonds are broken and formed.
 - What are the pros and cons (including environmental and socio-economic impacts) of using this chemistry to generate the energy?

Process and Analyze

1. What is the source of and use for the energy you chose?
2. Were you able to determine the answers to all of your questions? If so, what are they? If not, why were you not able to? What information was lacking?

Conclude and Communicate

2. Summarize the advantages and disadvantages of using this chemical reaction and energy source.
3. Combine your group's findings with those of the class. As a class, generate a presentation that summarizes all of your findings.